ABHIRAM REDDY TADI

CAD Technician

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Full UK right to work (Graduate visa) | Available immediately

PROFESSIONAL SUMMARY

MSc Advanced Product Design Engineering & Manufacturing graduate (Kingston University London) and Certified SolidWorks Associate (CSWA). I produce accurate 3D models, assemblies and BS 8888 manufacturing drawings with dimensions, tolerances and BOMs, and design with manufacture in mind (DFM/DFMA). Strong attention to detail; keen to work closely with design and production teams.

KEY SKILLS

CAD software: SolidWorks (CSWA): 3D modelling, parts, assemblies, configurations, 2D drawings, BOMs, mass properties; AutoCAD (basic); Siemens NX (basic); ANSYS DesignModeler
Technical drawings: 2D and 3D CAD drawings, detailed manufacturing and production drawings, general arrangement (GA) drawings, BS 8888, GD&T, dimensioning and tolerancing, bills of materials (BOM)
Manufacturing: Design for manufacture and assembly (DFM/DFMA), design changes and modifications, prototyping and fit testing, FDM 3D printing (Bambu Lab A1, PLA/PETG), CAM simulation in Siemens NX
Measurement & quality: Reverse engineering from physical parts, caliper measurement, CMM inspection, 3D laser scanning, quality management
Other: ANSYS Workbench/CFX (CFD), basic FEA, Microsoft Office, technical report writing

PROJECTS

CAD Design Project (self-directed), London | 2026 – Present
Reverse engineering and technical drawings: Victorinox Classic SD multi-tool, SolidWorks

- Reverse-engineered a 58 mm, seven-function multi-tool into a 16-part parametric SolidWorks 3D model and assembly, with no manufacturer CAD or drawings, working from the physical product, caliper measurements and reference photos.
- Prepared BS 8888 technical drawings from the 3D model: a general arrangement drawing of the full assembly with balloons and a bill of materials, plus dimensioned and toleranced detail drawings of four key parts.
- Modelled complex geometry with lofts, sweeps and patterns (backsprings, blade profiles, pressed liner recesses) and used interference detection to find and fix fit problems between parts.
- Made three full-scale 3D-printed form-and-fit replicas (Bambu Lab A1, PLA/PETG), checking each against the original for proportions and pivot spacing and revising the CAD model between prints to correct clearances and tolerances. Combined the three-part scissors into one print-in-place body with designed clearance gaps.
- Assigned engineering materials (X55CrMo14 steel, AISI 304 stainless, nickel silver) for mass properties and rendered the assembly in SolidWorks Visualize; renders, drawings and replica photos are on my portfolio.

EDUCATION

MSc Advanced Product Design Engineering & Manufacturing | 2024 – 2025
Kingston University London

- Modules: Computer Integrated Product Development; Advanced CAD/CAM Systems; Quality Management.
- Lab work: CMM (coordinate measuring machine) inspection, 3D laser scanning, metrology and materials testing.
- Dissertation: CFD optimisation of rooftop wind turbine layouts on a 200 m building in ANSYS (2.2-million-element mesh). The simulations predicted about 32% higher power output for a staggered layout than a grid layout.

BTech Mechanical Engineering | 2019 – 2023
SRM University, Chennai, India
Engineering design, machine design, manufacturing processes, materials science, fluid mechanics.

CERTIFICATIONS & DEVELOPMENT

- Certified SolidWorks Associate (CSWA), Dassault Systèmes, July 2026
- Preparing for the Certified SolidWorks Professional (CSWP) exam, with regular SolidWorks speedmodelling practice